



Evaluating the usability of open source software tools in e-learning: a systematic mapping study and case study

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Abstract

Adopting open-source software (OSS) tools in modern education underscores the critical need for usability research to enhance learning experiences. This study has two objectives: to identify the usability techniques used in e-learning OSS projects and to evaluate the usability of an OSS e-learning tool, the LMS Desktop Assistant. A systematic mapping study (SMS) addressed the first objective by reviewing four significant databases: IEEE Xplore, Web of Science, ACM Digital Library, and Scopus, covering 2005 to 2024. This search resulted in 53 primary studies, revealing usability techniques such as questionnaires, heuristic evaluation, direct observation, surveys, interviews, and feedback mechanisms. The second objective involved using the Direct Observation and Test Questionnaire to evaluate the usability of the LMS Desktop Assistant, following an iterative approach incorporating student feedback to refine the tool's interface and functionality. The study also identified the need to explore personalization and cultural considerations in OSS e-learning environments. The SMS revealed a growing interest in techniques to improve user experience within educational OSS platforms, offering insights for more inclusive and effective e-learning environments. The LMS Desktop Assistant evaluation also highlighted key usability improvements, reinforcing the importance of user-centered design in OSS. This research provides practical guidance for developing more user-friendly OSS platforms that contribute to advancing educational technologies.

Keywords Open-source software · Usability techniques · E-learning · Educational institutions · Systematic research · User testing

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Introduction

Open-source software (OSS) promotes unrestricted freedom to utilize, study, modify, and distribute software. This philosophy, championed by pioneers like Stallman and the Free Software Foundation, has seen significant growth (Free Software Foundation; Viñar, 2022). OSS offers numerous benefits, including collaborative innovation and enhanced security due to its transparency, allowing the community to promptly detect and rectify any flaws or vulnerabilities (Piedrahita, 2005; Viñar, 2022). However, OSS also presents challenges, such as the need for centralized support and the occasional lack of transparency in business models, which can deter investment. Despite these challenges, OSS influences the technology industry significantly (Casanova & García, 2021). The expansion of OSS and advancements in e-learning have transformed the educational landscape, making education more accessible, flexible, and inclusive (Barberà-Gregori & Suárez-Guerrero, 2021; Gros Salvat, 2018). E-learning tools, which allow access to educational content anytime and anywhere, have been extensively integrated into academic programs (Camacho Terán, 2019; Mariaca Garron et al., 2021). These tools foster interactive and collaborative learning and enhance students' preparedness for higher education and the workforce (Da Silva & Núñez, 2021). Usability in digital tools is vital to ensuring optimal experiences, facilitating learning, and minimizing student frustration (Aguaded Gómez & Fandos Igado, 2009; Del Prete & Cabero Almenara, 2019; Kumar & Mohite, 2018). In OSS, usability is addressed collectively. However, lacking a dedicated team can impede improvements (Almanza, 2021; Jiménez Galán et al., 2021). Although traditionally focused on functionality, the importance of usability in OSS for broader adoption is gaining recognition (Paniagua et al., 2020). Usability has become an area of increasing interest in OSS e-learning projects, as it plays a crucial role in ensuring that applications are user-friendly, accessible, and meet users' expectations (Alshamari & Althobaiti, 2024). The collaborative nature of OSS allows for valuable user information to be gathered, but this only sometimes leads to systematic improvements (Alarcón-Aldana et al., 2014; Llerena et al., 2019). Projects prioritizing usability often have higher adoption rates and receive positive feedback (Area-Moreira & Adell, 2021).

Conversely, an educational tool with poor usability can lead to student dropouts, distractions, and learning barriers (Cough Novel, 2021). Learners may become frustrated and discontinue using a tool with an inadequate user interface (Angulo-Armenta et al., 2021). Therefore, investing in usability is not just a matter of design but also a pedagogical strategy (Area-Moreira & Adell, 2021). Educational institutions must acknowledge that the quality of their digital tools directly influences the quality of their education (Area-Moreira & Adell, 2021). Usability in OSS e-learning projects has sparked a growing interest in recent research (Cocunubo-Suárez et al., 2018). Several studies have examined the implementation and effectiveness of usability techniques in these educational platforms, highlighting several significant findings (Cocunubo-Suárez et al., 2018). However, projects focused on advanced

features sometimes need to pay more attention to essential aspects of design and user experience (Cocunubo-Suárez et al., 2018; Couoh Novelo, 2021).

Good usability practices provide more intuitive and engaging learning experiences (Del Prete & Cabero Almenara, 2019), encourage active student participation (Flores-Fernández & Riquelme, 2022), and are more flexible and adaptable to users' and institutions' changing needs (García-Ruiz et al., 2018). However, maintaining consistency and ease of use in decentralized development environments is complex (Flores-Fernández & Riquelme, 2022), and balancing rapid innovation with stability and ease of use is a recurring concern (Flores-Fernández & Riquelme, 2022). Furthermore, practical usability in OSS projects necessitates continuous and active community collaboration (2021), which is not always easy. Some areas have been minimally explored, such as the cultural adaptability of interfaces in e-learning OSS projects (Espinosa-Izquierdo et al., 2021; Guédez, 2005; Salazar et al., 2017) and personalization based on individual learning styles and needs (Álvarez-Torres et al., 2021). A direct link between usability techniques and student learning outcomes is also needed (Torres-Lara et al., 2021; Zambrano, 2007), this context requires a centralized and detailed resource that thoroughly addresses usability evaluation in e-learning OSS projects and their problems. Furthermore, cultural adaptability and personalization in e-learning OSS require more research (Angulo-Armenta et al., 2021; Couoh Novelo, 2021). Traditional usability techniques may not be entirely applicable in these unique contexts, necessitating more dynamic and adaptive approaches (Ceballos Rincón et al., 2019).

This research aimed to identify the usability techniques applied in OSS projects for e-learning and to synthesize the usability problems by proposing improvements. It also aimed to evaluate the usability of the LMS Desktop Assistant tool. A Systematic Mapping Study (SMS) was conducted to achieve the first objective, reviewing publications from the IEEE Xplore, Web of Science, ACM Digital Library, and Scopus databases, covering the period between 2005 and 2024. The main findings highlighted various usability techniques, including questionnaires, heuristic evaluation, direct observation, surveys, interviews, and feedback. We evaluated the usability of the LMS Desktop Assistant tool by applying two techniques: Direct Observation and Test Questionnaire for the second objective. Our contribution maps the primary studies on usability evaluation in e-learning OSS projects and suggests improvements based on the identified problems. It is argued that the simplicity of the interface is crucial for its adoption at the secondary level. However, it must be balanced with the functionality necessary for an enriching educational experience. An iterative and empirical approach was observed in usability evaluations with real students. The importance of further research on personalization and cultural considerations was also underscored.

The rest of this document is organized as follows: Sect. "Related work" describes the related work, Sect. "Systematic mapping study" describes the SMS conducted, Sect. "Results" presents the research results, Sect. "Discussion" discusses the results, Sect. "Validity threats" describes the threats to validity, and Sect. "Conclusions and future work" describes the conclusions and future work.

Related work

The field of education has undergone a significant transformation with the introduction and evolution of OSS and e-learning. These developments have opened new avenues for learning and teaching, presenting vital opportunities and challenges to understand and address. The literature on OSS in education, e-learning integration, and usability principles in educational tools offers a rich foundation for exploring these transformations. Fitzgerald's research (2006) and Crowston et al. (2012) reveal how OSS promotes collaborative learning and sustainable strategies, emphasizing the need for robust communities and adaptive approaches. Effective e-learning integration, according to Conole and Alevizou (2010), requires considering pedagogical and technological contexts to enhance the educational experience. In addition, according to Hornbæk (2006) and, Zaharias and Poylymenakou (2009), usability is essential for the effective adoption of e-learning tools. User-centered design and intuitive interfaces are crucial for successful interaction and engagement, highlighting the importance of feedback and iterative design. The present research on usability techniques applied in e-learning OSS projects used in Educational Institutions has been conducted based on existing literature, focusing on three critical areas that form the backbone of the study:

Evolution and scope of OSS in education

OSS has transformed education by providing accessible and adaptable tools. Recent research has highlighted how OSS fosters collaboration and innovation in educational settings. For example, Hilton (2016) examines how open educational resources can enhance learning and reduce costs. In addition, research by Fitzgerald et al. (2005) highlights the importance of the community in developing and maintaining OSS projects, resulting in a wide diversity of tools and resources. However, as noted by Watson et al. (2008), persistent challenges include the long-term sustainability of OSS projects and the need for more structured support.

Principles of usability in OSS e-learning tools

Dawood et al. (2019) study addresses the analysis and mapping of usability in OSS to promote the sustainability of this type of project. Through a systematic review, 53 relevant studies were identified and classified into four main categories: (i) OSS usability improvement, (ii) OSS usability analysis, (iii) OSS usability evaluation, and (iv) OSS selection and adoption. This work provides a structured taxonomy highlighting opportunities and gaps in OSS usability research, facilitating the identification of future directions for researchers interested in this field. Usability is crucial to the effectiveness of e-learning OSS tools. The study of Ardito et al. (2006) on usability principles in educational applications highlights the need for intuitive and adaptive interfaces. Nielsen (1989) also emphasizes the importance of simplicity and clarity in designing interfaces to enhance the learning experience.

In addition, Preece et al. (1994) point out that user feedback is vital for iterative design and continuous improvement of e-learning tools. Despite this knowledge, as Dillenbourg (2016) points out, there are still significant challenges in the effective implementation of usability practices in e-learning OSS projects, primarily due to the decentralized nature of OSS development. To the best of our knowledge, no work specifically investigates usability techniques applied in e-learning OSS projects, compiles the usability problems of this type of project, or makes improvement proposals for these problems. Therefore, more efforts are needed in the usability of OSS e-learning applications. For this reason, our research seeks to determine which usability techniques can be adequate to improve usability in OSS projects. In addition, we identify usability problems in this type of project and propose improvements.

Integration of e-learning in educational environments

E-learning has been increasingly integrated into secondary education, leveraging OSS platforms to deliver personalized learning experiences. Etherington (2008) argues that e-learning enhances accessibility and individualization of learning, allowing students to learn at their own pace. In their study, Greenhow and Robelia (2009) highlight how e-learning tools promote collaboration and student engagement. However, Selwyn's research (2010) indicates that effective e-learning integration requires careful consideration of pedagogical and technological contexts.

Systematic mapping study

According to Kitchenham and Charters (2007), an SMS allows a literature review on an area of interest that aims to answer one or more research questions. The questions posed for our research are as follows: (RQ1) What usability techniques are applied in e-learning OSS projects used in educational institutions? (RQ2) What are the usability problems of e-learning OSS projects? The process for performing an SMS reported by Kitchenham and Charters (2007) has three steps: (i) review planning, (ii) strategic search, and (iii) study selection. This process started with the identification of keywords and the search string. For this, we first read some initial studies that answered the research question, obtaining the technical vocabulary of the study area. After combining the opinions of two experts in OSS and e-learning, we opted for the following search string: (*"open source" OR "free software"*) AND (*usability OR "user experience"*) AND (*elearning OR e-Learning OR "electronic learning"*).

The search was conducted in four major databases: Scopus, IEEE Xplore, ACM Digital Library, and Web of Science, and the search period spanned from 2005 to 2024. The inclusion criteria used to determine the primary studies are: The study reports the use of usability techniques in e-learning OSS projects AND the paper is written in English AND its publication date is equal to or greater than 2004. The exclusion criteria used were as follows: The study does not present aspects

Table 1 Search field per database

BBDD	Search fields
IEEE Xplore	Abstract
Web of Science	Topic
ACM Digital Library	Abstract
Scopus	Title-ABS-Key

of usability techniques in e-learning OSS projects OR the study is a short paper, abstract or editorial. Table 1 shows the search field used for each of the selected databases.

The process for selecting the primary studies is explained below. First, the string to be used and the search fields for each database were defined (see Table 1). Subsequently, each database was searched. The resulting set of studies was referred to as Studies Found. Second, duplicate studies were removed from the Studies Found pool between each DB and all databases. When duplicate studies were found, the first occurrence of the study was retained, and the others were removed. Third, the studies were reviewed using the title, abstract, and keyword selection criteria. Fourth, the selection criteria were applied after reading and analyzing each of the studies belonging to this last group. Two experts in OSS and e-learning cross-checked the results of this selection, and any disagreements were discussed and resolved in meetings with the research team. After this process was completed, the Primary Studies were obtained. Finally, the search performed by SMS resulted in a total of 53 primary studies (PS). The references of the PS can be consulted in Appendix A.

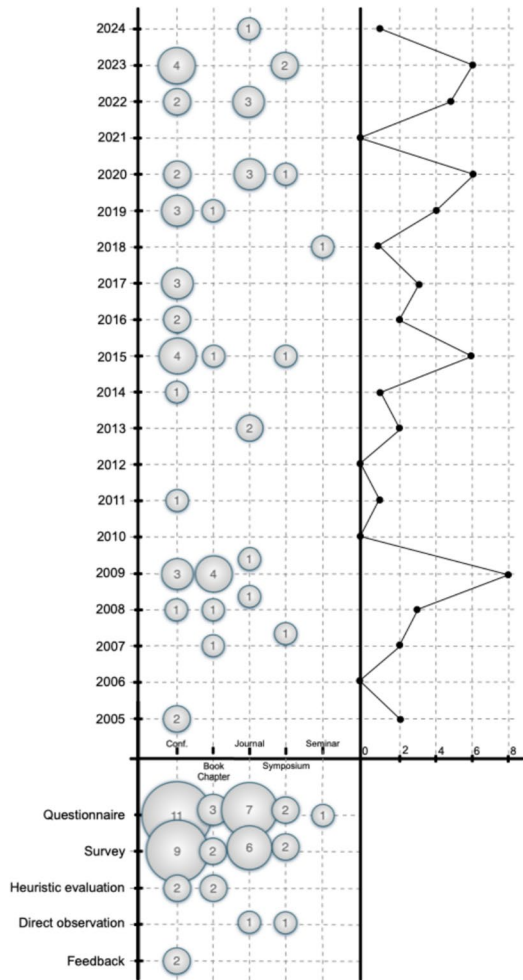
Results

This section reports the results obtained through our research. In subSect. "[SMS results](#)", we present the SMS findings that identify the OSS projects evaluated and the usability techniques used to perform the evaluation. SubSect. "[Usability problems identified in OSS projects and proposals for improvement](#)" summarizes the usability problems identified in the primary studies and proposes improvements to these problems. Finally, subSect. "[Usability evaluation of the LMS desktop assistant project](#)" reports the usability evaluation results of LMS Desktop Assistant. The usability evaluation of the tool is performed using the Direct Observation and Test Questionnaire techniques. Applying usability evaluation techniques highlights the challenges and opportunities in implementing usability practices and usability techniques reported in our SMS.

SMS results

This section presents the SMS results, divided into two parts: (i) A general overview of the primary studies and (ii) the answer to the research question posed in the

Fig. 1 Mapping for the primary study distribution between the different usability techniques along the year and type of publication



present investigation. Figure 1 summarizes the first part as a synthetic overview of the 53 primary studies identified.

The first (left side) displays two XY scatter plots (top and bottom), which show bubbles at the intersections of the categories: Type-year of publication and type-most used usability techniques. The publication types are conferences, chapter books, journals, symposiums, and seminars. The number of primary studies that have been classified in each category determines the size of the bubbles. As can be seen, the primary studies have been published primarily in conferences (49.1%) and journals (20.8%). The second area (upper right side) of Fig. 1 shows the number of primary studies published by year.

As can be seen, interest in evaluating e-learning OSS tools has changed over the search period. However, there are four periods where growth in the number of publications is noticeable: 2007–2009, 2014–2015, 2018–2020, and 2022–2023.

From 2018 to 2023, interest has grown, except for the abrupt drop in publications in 2021, perhaps due to the COVID-19 pandemic. The lower part of Fig. 1 displays a bubble chart showing the number of primary studies classified by the most used usability techniques in e-learning platforms. The methods included are Questionnaires, Surveys, Heuristic Evaluation, Direct Observation, and Feedback. The size of the bubbles represents the number of primary studies that applied each technique. Questionnaires and Surveys are the most frequently used techniques, with significant clustering of studies in these categories, especially for Questionnaires (24 studies) and Surveys (19 studies). The following is the response to RQ1 based on analyzing the primary studies identified in our SMS.

The **Questionnaire** technique is widely used across various fields to gather data on usability and user experience (Mayhew, 1999). In education, questionnaires have been used to evaluate the usability of OSS projects such as Moodle, which educational institutions extensively utilize globally to improve user interactions (Jingga & Sunindyo, 2020; Jókai, 2009; Kakasevski et al., 2008; Lavidas et al., 2023; McGrath, 2005; Sheshasaayee, & Bee, 2017; Ternauciuc & Vasii, 2015). In music education, the Troubadour platform enhances aural skills, using questionnaires to assess impacts on student learning in music theory (Pesek et al., 2020). Similarly, in computer science education, questionnaires assess tools like EvscApp, an educational escape game, and Virtual Computing Lab (VCL), a programming system, enhancing their usability and effectiveness (Kabimbi Ngoy et al., 2023; Li et al., 2009). Questionnaires in the FOSS web programming course provided crucial feedback to optimize its usability (Morelli & De Lanerolle, 2009).

Questionnaires in educational games like those on the CHAMILO platform are vital for assessing their appeal and functionality, proving essential in e-learning evaluations (Dinimaharawati et al., 2018). Similarly, integrating the instructional management system–learning design (IMS-LD) into e-learning tools has been studied using questionnaires to gauge usability, a crucial aspect of user experience (De Oliveira & Gomes, 2015). Questionnaires have also been vital in language learning evaluations, notably in the You Tutors system using the YouTube API, where they have supported enhanced teaching strategies validated by usability tests with English learners using questionnaires with a 5-point Likert scale (Nakajima & Ono, 2015). In mobile learning environments, such as Progressive Web Apps (PWAs), quizzes have been essential to obtain immediate feedback on user experience. The authors emphasize the usefulness of PWAs in promoting interactive learning by using questionnaires to get immediate feedback when evaluating these applications (Maskeliunas et al., 2023). OpenPose, an OSS tool, uses questionnaires to monitor student engagement and assess motivation and academic outcomes under ISO-9241-11 standards (Sabuncuoglu & Sezgin, 2023).

Additionally, the OSS CGDemo tool, designed for computer graphics education, is evaluated through questionnaires to determine its effectiveness in teaching algorithms and collecting user feedback on usability (Kumar & Zhang, 2020). The W3C Accessible Rich Internet Applications (ARIA) suite, an OSS educational platform, promotes innovation in teaching and learning, emphasizing virtual environment setup, course design, and training strategies. The evaluation results of the suite highlight its usability, such as its intuitive interface, simple navigation,

and attractive design, which were validated through a questionnaire (Buzzi et al., 2009). On the other hand, the Widgets for Inclusive Distributed Environment (WIDE) project uses participatory design and agile development to create open-source widgets aiding students with disabilities and enhancing e-learning inclusivity (Gkatzidou et al., 2011). Based on questionnaires, OSS Learning Management Systems (LMS) like Open eclass and Clavis were evaluated to improve usability (Hermann et al., 2019; Papantoni et al., 2019). Specifically, Open eclass focuses on e-learning effectiveness, web influence, and critical LMS features with usability testing (Papantoni et al., 2019), while Clavis examines learner interactions within learning environments (Hermann et al., 2019). In the Santos and Boticario (2008) study, the dotLRN recommendation model for e-learning was explored, focusing on usability, and integrated into an OSS-compliant learning management system. Usability was assessed via questionnaires to enhance user experience (Santos & Boticario, 2008). Similarly, the EX-COFALE e-learning system for nursing was evaluated for ease of use through targeted questionnaires (Hatzilygeroudis et al., 2007). Additionally, the Continuing Professional Development (CPD) module for dentistry training, developed by distance learning experts, was assessed for content and usability with a 31-question questionnaire (Kossioni et al., 2013).

The **Survey** technique effectively gathers data on user satisfaction (Ferré, 2005). Moodle's usability was assessed via surveys in educational settings, highlighting necessary (Cadena-Vela et al., 2017; Ivanović et al., 2013; Saleh et al., 2022; Sheshasaayee & Bee, 2017). Similarly, the Full Gamification Programming Education (FGPE) platform, designed for programming with gamification, motivates students through challenges; its usability was evaluated using surveys (Paiva et al., 2022). Another e-learning platform leveraging the Content Management Interoperability Services (CMIS) API to integrate repositories scattered on the Internet, facilitating access to educational resources, has also been evaluated. The usability of this e-learning platform has been assessed through surveys, suggesting its potential for widespread adoption (Favario et al., 2015). In LMSs for e-learning, usability surveys have been used to enhance the student experience via personalized recommendations on the dotLRN platform (Santos & Boticario, 2009). The usability of the Medical Education Resources Gateway (MERG) e-learning tool was also evaluated. MERG is a suite for exploring competencies and educational resources. The user experience was examined through surveys to detect shortcomings in the content offered (Bhupatiraju et al., 2008). An Electronic Health Record (EHR) system course was adapted using OSS tools in healthcare. The authors evaluate the usability of EHR using surveys to assess the effectiveness of pedagogical approaches (Zhang & Shahriar, 2019). For MOOCs on platforms like edX and Coursera, challenges in managing large student interaction databases were analyzed. ELAT, an OSS tool, was developed to analyze MOOC data, and its usability was evaluated through surveys to optimize user feedback interpretation (Torre et al., 2020). The SUS survey has been used to assess the usability of Moodle and Open edX (Liapis et al., 2022). Ouh et al. (2022) introduced the Interactive Tutorial Software System (ITSS) to enhance programming interactivity by allowing browser action recording and playback. The SUS survey evaluated ITSS usability with university students (Ouh et al., 2022).

In mathematics, two OSS tools have been evaluated (Cao & Grabchak, 2023; Franceschini et al., 2019). WeBWork, used for discrete mathematics, has proven effective in generating and self-assessing problems to enhance student performance, supported by student surveys (Cao & Grabchak, 2023). The Inequality project, comparable in effectiveness to traditional methods, aims at improving problem-solving in online STEM education. Its usability, particularly in high school settings, has shown high satisfaction levels, according to surveys (Franceschini et al., 2019). In Technology and Information Systems education, the Technology Acceptance Model (TAM) has been applied to educational technologies like e-Portfolios, assessing usability through user surveys (McCrath, 2005). In Software Engineering, the FlossSearch.Edu project has been noted for facilitating the search for educational resources. The usability evaluation of this project was also based on the TAM model. The TAM model, through surveys, allows users to measure their perceptions of ease of use and usefulness (Lessa et al., 2020). Also, within the context of computing, in an introductory course on Cybersecurity, gamified learning has been integrated through Capture the Flag (CTF) challenges on the OSS CTF platform. This platform uses surveys as a usability technique to evaluate and improve the educational experience (Kaplan et al., 2022).

Similarly, the OSS GINI project enhanced computer network study via virtual micro-internets. Surveys have been used to assess their usability, confirming their effectiveness in computer networking courses (Maheswaran et al., 2009). Similarly, BitFit, an online Introduction to Programming course tool, was evaluated using usability surveys. It facilitated learning, collected data, and supported at-risk users, providing vital feedback (Estey & Coady, 2016). UBUMonitor, an open-source desktop application for monitoring student activity on Moodle, was also evaluated through surveys with educators. These surveys revealed high levels of user satisfaction, scoring above average in clarity, efficiency, and reliability, supporting its effective use in educational settings (Marticorena-Sánchez et al., 2024).

The **Focus Group** technique involves a series of meetings involving end users and usability experts. In these meetings, user issues and concerns related to the user interface are discussed (Nielsen, 1993). This technique was used to collect ideas for games to enrich the educational environment of EvscApp, an OSS project designed for Computer Science education. Using the Focus Group technique, participants complete their opinions about their experiences and preferences regarding the EvscApp project (Kabimbi Ngoy et al., 2023).

The **Direct Observation** technique enables direct user observation while they perform tasks, noting their behavior (Nielsen, 1993). This technique evaluated two OSS tools, Troubadour (Pesek et al., 2020) and Twine (Iwasaki & Hardin, 2023). Troubadour enhances auditory music theory learning via gamification; its evaluation utilized usability techniques like Direct Observation to optimize user experience (Pesek et al., 2020). Twine, an interactive programming tool, was assessed at a camp for children aged 8–15, confirming its educational simplicity and preference (Iwasaki & Hardin, 2023).

The **Heuristic Evaluation** technique involves examining an interface to identify its strengths and weaknesses (Nielsen, 1993). This technique assessed WebGoat's usability, an OSS e-learning system offering web application security training,

revealing frequent usability issues (Amin & Saeed, 2015). With this technique, the authors analyzed how the interactive features of e-learning OSS platforms, such as Moodle, ILIAS, Sakai, ATutor, Claroline, Dokeos, and OLAT, influence the user experience. Particularly, Moodle and Dokeos excelled in creating engaging, functional environments (Al-Muaythir et al., 2014). The rising popularity of LMSs like Moodle is attributed to their reliability, which is crucially supported by open standards and heuristic evaluations (Desai et al., 2016). Martin et al. (2007) studied platforms aLFanet, EU4AL, and ALPE, enhancing educational collaboration and accessibility, including for special needs, through heuristic evaluation. Additionally, online LMSs improve user experiences with personalized recommendations, as seen in dotLRN's usability assessment (Santos & Boticario, 2009).

The **Forums** technique allows the publication of messages and questions in spaces on the web. These newsgroups cover programming languages, software tools, or tasks (Constantine & Lockwood, 1999). The study by Lichtner et al. (2009) investigated using APOSDLE, a knowledge management tool for self-directed learning in work environments, through a Forum. This forum, created with the OSS phpBB software, recorded the use and usability issues of the tool in several European organizations. Its flexibility facilitated adaptation to users' needs, playing an essential role in evaluating and improving APOSDLE (Lichtner et al., 2009).

The **Prototypes** are reduced versions of the complete system, which may limit the number of functionalities or reduce the level of functionality of options that appear to work but do not do anything (Nielsen, 1993). In developing the ILIAS LMS, usability and interface design principles were incorporated. To overcome usability challenges in OSS projects, Kitchen Sink, an innovative solution combining specifications, documentation, illustrations, design, and automated testing, was implemented. Rapid iteration of prototypes and the centrality of usability in development were critical aspects of this process (Masson et al., 2017).

Interviews are a technique for obtaining users' subjective reactions (Nielsen, 1993). In the context of the OSS e-learning platform Troubadour, designed for auditory training in music theory using gamification, interviews were conducted with teachers and students. These interviews defined the desired functionalities and understood students' engagement with mobile devices, especially games and music applications (Pesek et al., 2020). User feedback can be collected by giving users specific e-mail addresses, news groups, or electronic bulletin boards to send complaints, change requests, or improvement requests (Nielsen, 1993).

In the primary studies, we identified two cases using the **Feedback** technique (Bonastre et al., 2005; Tick, 2009). The first case corresponds to E-dap, an e-learning tool that assesses individual students' progress by adjusting the difficulty of questions in real-time in exams based on their previous performance. In this tool, the importance of performing usability evaluations and incorporating user feedback to measure the effectiveness and efficiency of the tool was highlighted (Bonastre et al., 2005). In the second case, the study's authors developed a military English course in Moodle, implemented through the Military English Learning System (MILES). They explored the perceptions of the usability of the material and the users' learning patterns. The feedback collected through Moodle and other analysis

tools identified areas for improvement in the course's content and structure, thus facilitating the updating and improvement of the system (Tick, 2009).

Scenarios are used for systems that will face substantial changes or when a novel application is proposed, and there needs to be more reliable data on task range and frequency distribution. This technique involves describing the system using usage scenarios and, if possible, performing them as if it were a play (Shneiderman, 1998). In the case of Moodle, usability testing was conducted with fifteen participants, including students and users of Korlantas, a special division of the Indonesian National Police, to assess their experiences with the platform. Acting as students within the developed e-learning environment, the participants ran four pre-determined scenarios. Then, they completed a survey about their experience, including the ease of learning the courseware. The Scenarios technique helped evaluate the system's usability and improve the user experience (Jingga & Sunindy, 2020).

In the **Laboratory Test**, participants are taken to a usability lab-type facility to perform benchmark tasks and other aspects of a usability test (Constantine & Lockwood, 1999). In a study on Moodle usability, shortcomings were identified, and solutions were offered after comparison with different platforms. The researchers highlighted the importance of regularly evaluating Moodle usability, proposing specific methodologies that included laboratory tests with regular users and heuristic analysis by usability experts. Evaluation through Laboratory Tests helped to identify areas for improvement in Moodle usability (Ternauciuc & Vasii, 2015).

The **Logging** technique consists of collecting all the activities performed by the user and integrating them with the system for later analysis. This technique is used to identify usage patterns and areas for improvement in the user interface and user experience (Paganelli & Paternò, 2002). The SIVA Suite, a tool that facilitates creating and managing hypervideos, was highlighted in the educational field. Log data on user interactions with the SIVA Player were collected and analyzed to evaluate its usability. This allowed us to identify usage patterns and areas for improvement in the user interface and user experience (Meixner et al., 2015).

In **Expert Review**, the reviewers are experts in the application or user interface domain. Depending on the review's focus, there are different types of (Shneiderman, 1998). In a detailed analysis of the usability of LMS tools, the following were evaluated: Moodle, ATutor, Eliademy, Forma LMS, and Dokeos. These tools are learning management systems that allow educators to create and manage online courses. Using the ISO/IEC 9126 quality assessment model, usability issues were identified in these LMS tools. Two experts evaluated the above tools based on established criteria, underlining their importance in today's digital education. The Expert Evaluation helped to identify usability problems in these systems and to improve their design and functionality (Başaran & Mohammed, 2020). Table 2 is composed of 3 columns: (i) usability technique, (ii) OSS project evaluated, and (iii) primary studies reporting the use of the usability technique.

Table 2 Usability techniques were identified in the primary studies

Usability technique	OSS project evaluated	Primary study
Questionnaire	Moodle	(Jingga & Sunindyo, 2020; Jókai, 2009; Kakasevski et al., 2008; Lavidas et al., 2023; McCrath, 2005; Sheshasaayee, & Bee, 2017; Ternauciuc & Vasiiu, 2015)
	Troubadour	(Pesek et al., 2020)
	EvscApp	(Kabimbi Ngoy et al., 2023)
	VCL	(Li et al., 2009)
	FOSS	(Morelli & De Lanerolle, 2009)
	CHAMILO	(Dinimaharawati et al., 2018)
	IMS-LD	(De Oliveira & Gomes, 2015)
	You Tutors	(Nakajima & Ono, 2015)
	PWAs	(Maskeliunas et al., 2023)
	OpenPose	(Sabuncuoglu & Sezgin, 2023)
	CGDemo	(Kumar & Zhang, 2020)
	ARIA	(Buzzi et al., 2009)
	WIDE	(Gkatzidou et al., 2011)
	Open eclass	(Papantoni et al., 2019)
	Clavis	(Hermann et al., 2019)
	dotLRN	(Santos & Boticario, 2008)
	EX-COFALE	(Hatzilygeroudis et al., 2007)
	CPD	(Kossioni et al., 2013)
Survey	Moodle	(Cadena-Vela et al., 2017; Ivanović et al., 2013; Saleh et al., 2022; Sheshasaayee & Bee, 2017)
	e-Portfolios	(McCrath, 2005)
	FGPE	(Paiva et al., 2022)
	e-learning platform for educational resources	(Favario et al., 2015)
	dotLRN	(Santos & Boticario, 2009)
	MERG	(Bhupatiraju et al., 2008)
	HER	(Zhang & Shahriar, 2019)
	ELAT	(Torre et al., 2020)
	Moodle, Open edX	(Liapis et al., 2022)
	ITSS	(Ouh et al., 2022)
	WeBWorK	(Cao & Grabchak, 2023)
	Inequality	(Franceschini et al., 2019)
	FlossSearch.Edu	(Lessa et al., 2020)
	CTFd	(Kaplan et al., 2022)
Focus Group	GINI	(Maheswaran et al., 2009)
	BitFit	(Estey & Coady, 2016)
	UBUMonitor	(Marticorena-Sánchez et al., 2024)
	EvscApp	(Kabimbi Ngoy et al., 2023)
Direct	Troubadour	(Pesek et al., 2020)
	Twine	(Iwasaki & Hardin, 2023)
Observation		

Table 2 (continued)

Usability technique	OSS project evaluated	Primary study
Heuristic Evaluation	dotLRN	(Santos & Boticario, 2009)
	WebGoat	(Amin & Saeed, 2015)
	Moodle, ILIAS, Sakai, ATutor, Claroline, Dokeos, OLAT	(Al-Muaythir et al., 2014)
	Moodle	(Desai et al., 2016)
Forum	aLFanet, EU4AL, ALPE	(Martin et al., 2007)
	APOSdle	(Lichtner et al., 2009)
Prototypes	ILIAS	(Masson et al., 2017)
Interviews	Troubadour	(Pesek et al., 2020)
Feedback	E-dap	(Bonastre et al., 2005)
	MILES	(Tick, 2009)
Scenarios	Moodle	(Jingga & Sunindyo, 2020)
Laboratory Test	Moodle	(Ternauciuc & Vasiiu, 2015)
Logging (recording of Usage)	SIVA	(Meixner et al., 2015)
Expert Review	Moodle, ATutor, Eliademy, Forma LMS, Dokeos	(Başaran & Mohammed, 2020)

Usability problems identified in OSS projects and proposals for improvement

Based on the analysis of the primary studies obtained from the SMS, this section identifies the usability problems of OSS e-learning projects and makes proposals for improving them. Table 3 summarizes these problems and the improvements identified. This analysis is fundamental to optimizing the usability of these educational platforms, thus offering more robust and efficient solutions that better meet users' needs in academic environments.

Usability evaluation of the LMS desktop assistant project

This section reports the results of applying the Direct Observation and Test Questionnaire techniques in the LMS Desktop Assistant project. This OSS project facilitates the management of educational documents, allowing file conversion and the generation of statistics, such as the percentage of access to academic resources. This improves teachers' efficiency and enriches students' learning experience. Its choice as a case study is particularly appropriate since it presents usability characteristics and challenges analogous to those identified in the SMS.

This allows a direct and relevant comparison with usability techniques applied in other OSS educational projects, facilitating the extrapolation and application of proven and practical improvements in a similar context. This case study selected

the Direct Observation and Test Questionnaire techniques as usability evaluation methods because they provide complementary insights: Direct Observation allows us to analyze how users interact with the OSS project in real-time, capturing spontaneous behaviors and natural reactions, while the Test Questionnaire collects users' structured feedback on their experience, helping to identify usability issues and areas for improvement.

This is crucial to identify usability problems that other methods might not detect. It also facilitates understanding the context of use, essential in educational environments. Direct Observation provides valuable insights to improve the interface and functionality of the software by capturing real-time interactions. At the same time, the Test Questionnaire allows participants to reflect on their experience and provide structured feedback on usability issues. Seven users participated in the usability evaluation of LMS Desktop Assistant, a number justified by (i) its ability to capture a variety of usability issues and (ii) its efficiency in terms of resources and time, allowing a detailed analysis. Studies show five users can uncover the most usability issues (Nielsen, 1993).

Furthermore, in the educational context, a manageable group of participants facilitates the coordination and effectiveness of testing sessions. Before this usability evaluation of LMS Desktop Assistant, contact was established via e-mail with its developers. This communication was aimed at establishing an effective collaboration to work on the source code of the OSS project. The primary purpose of this evaluation is to analyze the usability of the OSS project to understand how it adapts to users' needs in the context of learning management. This initial phase of the process is fundamental to evaluate the efficiency and effectiveness of the tool before making any possible improvements that may influence its usability. To initiate the usability tests, an e-mail invitation was sent to seven specific users, colleagues, or acquaintances of the researchers and, simultaneously, students from the university where one of the researchers works. This invitation included the date and time scheduled for the pilot test. In addition, two links were provided.

The first was to join the WhatsApp group designated for the test, while the second had to meet for the evaluation video call. Two additional files were also included in the invitation. One document contained the tasks users were to perform during the test; another included a structured feedback questionnaire (see <https://lc.cx/rHJ2wm>), and the other included the consent form they were to sign to participate in the study (see <https://lc.cx/xYybrv>). The opening videoconference began by thanking the participants for their collaboration. A concise introduction to the project to be evaluated was given so that users would have a clear idea of the project's scope before conducting the usability test. Before the evaluation, participants also signed a consent form authorizing the capture of images on video and using their pictures in the research context.

At the end of each evaluation, participants completed the questionnaire, which allowed them to freely express their opinions on system usage. This instrument facilitated the collection of valuable feedback, providing insights into usability experiences and potential areas for improvement.

Below are two specific tasks: (i) selecting a file and (ii) converting a document to PDF format using the application. These tasks provide clear and detailed instructions

that facilitate understanding and executing the required actions in the LMS Desktop Assistant tool.

Table 4 provides a detailed description of the task's users were required to perform using the LMS Desktop Assistant tool in its original state.

Table 5 presents an example of the document used to collect incidents while applying the Direct Observation technique in the LMS Desktop Assistant tool context. It details Task 2 that the users carried out. The additional comments provide valuable information that can be crucial for identifying areas requiring improvement or adjustment in the OSS project.

Table 6 presents a compilation of problems identified by users during the Direct Observation. Each entry describes a specific issue, the users who experienced that problem (we assigned an identifier to each user who participated in the evaluation), and the suggestions for improvement proposed by the users.

Based on the results obtained during the application of the Direct Observation technique, we identified that the main criticisms of LMS Desktop Assistant were its interface, which needed to be updated, made more straightforward, and limited options. Many users also indicated the need for an integrated help manual and a quick guide for new users.

In addition, the limitation in file conversion was mentioned; for example, it allows the conversion of Word documents to PDF format, leaving out other formats, such as PowerPoint or Excel. It was also suggested that the design's consistency and the interface's adaptability on different devices be improved, and a search bar and more explicit icons should be added. Finally, a Spanish version and customizations to the interface design were requested to make the tool more attractive and intuitive.

Table 7 summarizes the results obtained through the questionnaire. This technique required participants to answer six open-ended questions based on their experience interacting with the system.

The usability evaluation of the LMS Desktop Assistant using Direct Observation and Test Questionnaire techniques provided valuable insights into the tool's strengths and areas for improvement. These methods allowed for a comprehensive assessment by capturing real-time user interactions and structured feedback, highlighting usability challenges that may not have been detected through a single technique alone.

The findings indicate that while the LMS Desktop Assistant facilitates document management and educational resource access, its usability could be enhanced through interface improvements, better navigation flow, and expanded file format compatibility. Participants emphasized the need for a help manual, a more intuitive UI, and outstanding customization options, including support for multiple languages. Additionally, the study revealed that more evident iconography, a search function, and improved task efficiency would significantly improve user experience.

Table 3 Usability problems by primary study and improvement proposals

Primary study	Usability problems/errors, shortcomings/inconveniences found in the primary studies	Proposals for improvement
(Bonastre et al., 2005; Hatzilygeroudis et al., 2007; Ivanović et al., 2013; Kakasevski et al., 2008; Lavidas et al., 2023; Martin et al., 2007; Sabuncuoglu & Sezgin, 2023; Santos & Boticario, 2008)	Not allow customization for users (e.g., it is not possible to change the user profile) Not have enough settings for customization Lack of customization according to individual needs	Provide customization options within the platform Redesign the interface to align with the real world and improve documentation Implement customization and adaptability options
(Bhupatiraju et al., 2008; Cadena-Vela et al., 2017; Hatzilygeroudis et al., 2007; Jókai, 2009; Kakasevski et al., 2008; Kaplan et al., 2022; Kossioni et al., 2013; Lessa et al., 2020; Li et al., 2009; Liapis et al., 2022; Maheswaran et al., 2009; Paiva et al., 2022; Pesek et al., 2020; Saleh et al., 2022; Santos & Boticario, 2008)	The interface is complicated and not intuitive, for all users, especially for less experts The interface is confusing due to the variety of modules The user interface is complex making navigation difficult The interface is unattractive	Redesign the user interface to be intuitive, easy to use and accessible Simplify and clarify the user interface by reducing on-screen elements Redesign navigation with clear menus and a search bar
(Başaran & Mohammed, 2020; Hatzilygeroudis et al., 2007; Jingga & Sunindyo, 2020; Meixner et al., 2015; Paiva et al., 2022)	Difficulty navigating and finding lessons The interface has too many functions Issues with navigation due to the different ways instructors organize content Navigation is confusing and unintuitive	Redesign the user interface to be intuitive and less feature saturated Establish standards on how the content of the work should be managed Improve the navigation structure with clear and direct menus for a smoother experience Improve the navigation structure in the search function
(Başaran & Mohammed, 2020; Bhupatiraju et al., 2008; Bonastre et al., 2005; Dinimaharawati et al., 2018; Franceschini et al., 2019; Ivanović et al., 2013; Kaplan et al., 2022; Kossioni et al., 2013; Kumar & Zhang, 2020; Liapis et al., 2022; Nakajima & Ono, 2015; Paiva et al., 2022; Sheshasayee, & Bee, 2017; Tick, 2009)	Lack of immediate, detailed, and visible feedback in evaluations and tasks Lack of real-time feedback Lack of immediate technical support and feedback mechanisms when students face problems	Implement more detailed and useful feedback functionality for assignments and exams in real time, which guides you about errors and how to correct them automatically Implement gamification systems that offer badges and automatic more interaction Implement more interactive, complete feedback systems and clear notifications

Table 3 (continued)

Primary study	Usability problems/errors, shortcomings/inconveniences found in the primary studies	Proposals for improvement
(Hatzilygeroudis et al., 2007; Ivanović et al., 2013; McCrath, 2005; Sabuncuoglu & Sezgin, 2023; Santos & Boticario, 2008)	The interfaces do not comply with accessibility guidelines Platform not fully accessible or customizable Lack of customization and accessibility options The system is not accessible to all users Lack of compliance with accessibility standards	Adapt interfaces to meet accessibility standards and offer advanced customization options Increase accessibility and allow customizations based on user needs
(Ternauciu & Vasiu, 2015)	The Moodle platform presents challenges in terms of accessibility and navigability Integrating external tools into Moodle is complicated	Improve accessibility by following WCAG guidelines Integrate detailed documentation to facilitate integrations with external tools
(De Oliveira & Gomes, 2015; Kabimbi Ngoy et al., 2023)	Insufficient technical support and limited documentation Insufficient documentation and support for users The interface may not be accessible to users with disabilities	Expand technical support and enrich online documentation with detailed resources Improve documentation and offer technical support Implement accessibility features in the interface Comply with accessibility standards to guarantee usability for everyone Implement WCAG accessibility techniques
(Bonastre et al., 2005; Cadena-Vela et al., 2017; Desai et al., 2016; Kabimbi Ngoy et al., 2023; Meixner et al., 2015; Nakajima & Ono, 2015; Papantoni et al., 2019; Tick, 2009)	The level of previous experience of the students is not considered	Adjust the course for beginners or differentiate learning paths based on skill level
(Morelli & De Lanerolle, 2009)	The impractical interface on mobile devices for older users	Redesign interface elements to improve accessibility and interaction on mobile
(Maskeliunas et al., 2023)	Gamification elements are poorly integrated with educational content	Ensure that gamified elements are related to educational progress and provide meaningful rewards
(Buzzi et al., 2009)	The system is not compatible with some screen readers, limiting its accessibility Difficulties with forms on older screen readers	Avoid tables for layout and improve the labeling of data tables Ensure clear labels for all types of browsers
(Gkatzidou et al., 2011)	The widget interface is complicated for some users The widgets do not fully comply with accessibility standards	Redesign widgets for a more intuitive and accessible interface Ensure that all widgets comply with international accessibility standards

Table 3 (continued)

Primary study	Usability problems/errors, shortcomings/inconveniences found in the primary studies	Proposals for improvement
(Hermann et al., 2019; Masson et al., 2017)	Non-intuitive interaction scenarios Usability problems on mobile devices, for example: Lack of options to customize the interface and functionalities according to individual user preferences Insufficient collection of feedback on usability Few usability experts were involved Excessive inclusion of functionalities	Redesign the mobile interface to improve usability on small screens and touch environments Simplify interaction flows, improve instructions, and help within the application Involve more experts and offer training to developers Review and eliminate unnecessary features
(Santos & Boticario, 2009)	The LMS interface and navigation are complicated It is difficult to understand and use the recommendations provided	Simplify the interface and improve the navigation structure for ease of use Make recommendations more accessible and clearer and offer usage guides
(Zhang & Shahriar, 2019)	Installation and configuration of EHR systems are difficult for students Lack of immediate support for online students during internships	Provide more detailed step-by-step instructions and demonstration videos Offer real-time support and online help sessions for practical tasks
(Torre et al., 2020)	Difficulty accessing and processing edX record data for users without programming skills Complex configurations that require technical knowledge	Provide video tutorials on how to process the data Include a wizard to facilitate configurations
(Ouh et al., 2022)	Traditional videos offer little interactivity, which can impact learning	Improve interactivity in ITSS by allowing actions such as highlighting text, navigating panels, and executing code in real-time
(Iwasaki & Hardin, 2023)	Loss of work due to errors during real-time collaboration Conflicts between students when editing simultaneously The initial setup is complicated for educators	Perform tests and fix bugs to avoid data loss Implement a conflict management system that notifies and resolves simultaneous issues Simplify group creation and management with guides and automated assistance

Table 3 (continued)

Primary study	Usability problems/errors, shortcomings/inconveniences found in the primary studies	Proposals for improvement
(Al-Muaythir et al., 2014; Amin & Saeed, 2015; Bonastre et al., 2005; Lichtner et al., 2009; Martin et al., 2007; Tick, 2009)	Too much information is presented at once The interface is complex and overwhelming	Break information into manageable segments to improve understanding Conduct specific usability tests to identify interface elements that may be confusing
(Başaran & Mohammed, 2020; Meixner et al., 2015)	The interface is not optimized for mobile devices	Improve mobile optimization and responsive design Optimize the interface for optimal performance and usability on mobile devices
(Marticorena-Sánchez et al., 2024)	The user interface could be simplified for better navigation Some users may find the clustering functionalities complex to configure	Simplify the interface for clearer navigation and ease of use Provide additional guidance or tutorials to assist in setting up clustering features and interpreting results

Discussion

Our investigation has yielded intriguing insights into the application of usability in e-learning OSS, a subject gradually gaining prominence in academic literature. Our research is novel in its exploration of this unique intersection, providing the first comprehensive assessment of usability techniques applied to e-learning OSS projects.

This aligns with Fitzgerald's (2006) observations, which highlight the evolution of OSS towards more commercially viable models, suggesting a likely increase in the adoption of usability practices as these platforms strive for broader acceptance in specific markets, including education. Our findings illuminate a significant discrepancy between the recommended usability practices and their implementation in e-learning OSS projects within educational institutions. This inconsistency can be ascribed to OSS development's decentralized and community-oriented nature, as indicated by the primary studies we reviewed.

Due to this unique development structure, the prioritization of features and enhancements may sometimes align with optimal usability practices. When juxtaposing our findings with existing literature, an opportunity emerges to formulate specific frameworks or guidelines that facilitate incorporating usability techniques into the development lifecycle of e-learning OSS tools. As noted in Fitzgerald's (2006) analysis, the absence of universally applicable guidelines necessitates additional research efforts to create accessible resources for developing innovative business models, product support strategies aligned with community values, and implementing measures to protect against intellectual property rights infringement. These resources aim to harmonize commercial value propositions with OSS community values, marking a fundamental shift in the software industry for developers and designers of these OSS platforms.

Interestingly, our literature analysis suggests that despite the growing interest in and adoption of e-learning OSS tools, the understanding and application of usability-related techniques within this context still need to be developed. This is congruent with the historical focus on functionality and security in OSS development, which has often left usability as a secondary consideration (Raza et al., 2012). Comparing our results with previous studies, a growing recognition of the importance of usability in e-learning tools becomes apparent, reflecting a transition towards more user-centered approaches in OSS development. However, the effective implementation of usability techniques in these projects requires more than a shift in the mindset of developers and designers. It also necessitates structural support and educational resources to guide their integration into OSS e-learning projects. Despite the novelty of this finding, these results may be limited by the constraints of resources and interface design expertise faced by some open-source projects (Flores-Cerna et al., 2021).

This analysis underlines the need for further research and development on usability applied to OSS e-learning tools. It points towards a future where functionality and usability are considered together in developing innovative and accessible educational solutions. Our analysis of the primary studies reveals that

Questionnaires and Surveys are the most commonly used usability techniques in e-learning OSS projects. This finding corroborates the results reported by Zaharias and Poylymenakou (2009), who identify these techniques as the most prevalent in the usability evaluation of e-learning tools. However, we observed a lower application of techniques such as Interviews, Forums, and Prototypes, which are valuable in the educational context as they promote active learning, collaboration, development of practical skills, and critical thinking (Garrison, 2011; Krug, 2014; Nielsen, 1993).

In analyzing the problems identified in the OSS e-learning projects of our SMS, we discovered significant usability issues, such as non-intuitive interfaces and a need for more support resources. These issues impede the adoption and effectiveness of these tools in educational environments. Similar challenges were identified when analyzing OSS LMS Desktop Assistant using the Direct Observation technique, particularly in initial configuration and navigation. These findings underscore the need for simpler interfaces and more accessible documentation, reflecting a general trend in the usability evaluation of OSS e-learning, where simplifying the interface emerges as a critical strategy to improve user adoption and satisfaction in educational environments.

In conclusion, our research underscores the importance of usability in developing and implementing e-learning OSS projects. Our findings highlight the need for further research and development in this area, pointing towards a future where functionality and usability are considered together in developing

Table 4 Tasks to be performed by users with *LMS Desktop Assistant*

Tasks to be performed in LMS Desktop Assistant
Task 1. Selection of a file - Start the LMS Desktop Assistant software - Select the Select Document option - Within the Browse ... we select the Word document we want to open - Once the Word document is selected, click on open - Verify that the name of the selected Word document is correct if it is displayed on the right side of the Select Document option Task 2. Convert a document to PDF - Start the LMS Desktop Assistant software - Select the Select Document option - Within the Browse ... we select the Word document we want to open - Select the Word document that we want to convert to PDF and click on open - Once selected our file to convert, we proceeded to look for the option PDF conversion - Once identified the above, we will look for the Convert to PDF button - Click on the button mentioned above and wait for the conversion process to complete - Once the conversion process is finished, we look in the folder where the file to convert was and verify that the document is in pdf format

Table 5 Document for collecting incidents while applying the direct observation technique

Tasks to be performed in LMS Desktop Assistant (observer's annotations)				
	Tasks to be Performed by the User	Did you?	Problems	Other comments
a	Start the LMS Desktop Assistant software	Yes	No	None
b	Select the Select Document option	Yes	No	None
c	Within the Browse ... we select the Word document we want to open	Yes	No	None
d	Select the Word document to convert to PDF and click on open	Yes	No	None
e	Once selected our file to convert, we proceeded to look for the option PDF conversion	Yes	No	They were annoyed by the fact that they could select the document to be converted in docx format. The user wanted other format types
f	Once identified the above, we will look for the Convert to PDF button	Yes	No	None
g	Click on the button mentioned above and wait for the conversion process to complete	Yes	No	None
h	Once the conversion process is finished, verify that the document is in pdf format	Yes	No	They were annoyed because the converted document is saved automatically and does not select a particular path to save it

innovative and accessible educational solutions. The insights gained from this study are instrumental in guiding future research and practice in e-learning OSS and contribute significantly to the existing body of knowledge.

Validity threats

In this section, we discuss the possible limitations that influence the findings of our SMS. The first threat to the validity of this study lies in the potential bias introduced during the selection of studies reporting the evaluation of e-learning OSS projects for educational institutions. Despite following rigorous criteria for selecting and classifying the primary studies, our personal preferences and experience could have influenced the process. To mitigate this risk, we resorted to peer review within the research team, although this mechanism does not eliminate the possibility of bias. A second threat concerns the coverage of the literature search. Despite using multiple relevant databases and carefully selected search terms to cover a broad spectrum of publications, it is plausible that some relevant studies still need to be identified. This could be due to limitations in the databases consulted, language restrictions, or the selection of search terms needed to capture the full diversity of terminology used in the field.

We recognize that the inclusion of publications in languages other than English or the exploration of additional databases could enrich the results of the SMS. Another important consideration is the rapid evolution of the e-learning and OSS field. OSS projects are dynamic, with updates and changes occurring continuously. This study reflects the state of the literature up to the time of the search. However, it is essential to consider that new contributions may provide additional or contradictory perspectives to those presented here. Finally, the interpretation of the results and the generalization of the findings present their limitations. Since this study focuses on e-learning OSS projects specific to educational institutions, the conclusions may not apply to other contexts or types of software. Future work must consider exploring different educational contexts and kinds of OSS projects to validate and extend the findings of this study. In summary, while the present SMS provides valuable insight into usability techniques applied in e-learning OSS projects in educational institutions, the above threats should be considered when interpreting the results and planning future research.

Conclusions and future work

In conclusion, our study reveals that usability techniques significantly influence the effectiveness of e-learning OSS projects. Through an SMS, this research identified and analyzed numerous primary studies, underscoring the crucial role of a user-friendly interface and an intuitive user experience in the success of e-learning tools. Our findings indicate that OSS projects can flourish despite

Table 6 Problems encountered by users during direct observation

Problem	User ID	Improvements proposed by users
The main problem is that the tool does not have a help manual and there are no help options in the interface	1,2	Include a help manual
The user interface is confusing, with too many options available	2	When entering for the first time, there should be a quick guide to the program's functions. This way, the user will be able to get the most out of the application
The interface is very outdated and lacks consistency in design and navigation. The layout of the interface elements can be confusing and unintuitive, making it difficult to find and access the necessary functions efficiently	3	The interface should include instructions on how to use the tool
Interface with saturation problems, which generates delays in the execution of tasks	4	The functions should be less complex for novice users
The tool is not in Spanish	5	Simplify the interface design by highlighting the important elements
The buttons are small. In general terms, the interface lacks color. The interface lacks icons to make the tool more interactive	5	The interface should adapt correctly to different screen sizes and devices
The interface lacks intuitive and user-friendly icons	6	Add a search bar in the center or at the top of the application
It only allows converting Word files to PDF	2,4,6,7	Use clear and consistent icons throughout the interface to improve the understanding of the functions

Table 7 Results from the LMS Desktop Assistant test questionnaire

Questions	Comments
What are the main problems you encountered in the LMS Desktop Assistant?	Participants identified several issues during system interaction, such as: - Lack of a user manual or guide for new users - Outdated and confusing interface - Unintuitive options and an overloaded screen layout - Limited file conversion formats (only Word to PDF) - Inability to select a custom save location for converted files
What are the main usability problems you found in the LMS Desktop Assistant?	Participants highlighted several usability issues, including: - Navigation is unintuitive and requires too many steps for basic actions - Lack of a search bar for quick access to specific functions - Unclear iconography and small buttons that hinder interaction - No support for multiple file formats in the conversion feature - No option to manually select the destination folder when saving files
Do you have any improvement proposals for interacting with the LMS Desktop Assistant?	Participants generally recommended: - Adding a user manual and a quick start guide upon first use - Improving the organization of interface elements to enhance navigation - Expanding compatibility to support file formats like PowerPoint and Excel - Including a search bar and improving button visibility - Implementing a system that allows users to choose the save location for converted files
Do you have any suggestions for improving the test?	Some participants suggested including a more diverse group of users to evaluate the system across different experience levels. They also proposed adding more tasks to test additional software functionalities
Do you have any criticism or complaints about the user interface?	Several participants considered the interface to be unattractive and lacking in dynamism. They also noted the absence of visual guidance to facilitate navigation and the need for more intuitive icons
How do you think the user interface (or part of it) could be redesigned?	In general, participants recommended a more modern and interactive interface redesign. They proposed improving menu organization, adding more representative icons, including customization options, and offering a Spanish version for greater accessibility

inherent challenges by implementing well-founded usability practices. Active communities and diverse user participation emerged as critical elements for usability enhancement in OSS. With their varied perspectives, these communities can contribute to developing more intuitive interfaces and accessible tools when their input is appropriately utilized. Hence, we advocate for closer collaboration between developers, educators, and their user base.

Furthermore, our study found that e-learning OSS projects prioritizing usability often experience higher adoption rates and better reception within their communities. This finding underlines the necessity to balance technical innovation and interface design's simplicity and clarity. We encourage development teams to persist with a user-centric approach to ensure the relevance and effectiveness of their tools. Our research also highlighted the need for more literature on cultural adaptability and personalization based on individual learning styles in e-learning OSS. Moreover, the limited studies addressing the usability of e-learning OSS applications in secondary education suggest a pressing need for intensified research efforts in this area.

In response to our first research question, we identified common usability problems in OSS e-learning tools, including user interface complexity and the need for adaptability. These issues impede navigation and autonomous learning, negatively impacting user experience and motivation. In response to the second research question, techniques such as questionnaires, interviews, and heuristic evaluations were used to enhance the usability of OSS e-learning tools. The most effective solutions incorporated user-centered designs, integrated tutorials, and real-time feedback mechanisms, facilitating initial use and promoting continuous and practical learning.

This study provides the first comprehensive assessment of usability techniques applied to e-learning OSS projects. These findings are particularly relevant for developing genuinely inclusive and effective e-learning tools, requiring further research to address these less-explored usability aspects. We underscore the link between usability techniques and student learning outcomes. Our findings suggest that a more significant investment in usability can enhance the learning experience and positively influence students' academic performance. Consequently, we urge educational institutions to implement robust usability techniques when selecting and deploying e-learning OSS tools.

Our SMS has furnished a broad overview of the interplay between usability and OSS in e-learning. The trends identified and recommendations proposed have the potential to steer future developments toward creating more effective and accessible learning environments. Lastly, the issues identified in this research underscore the need for intuitive interfaces and user experiences aligned with the profile of the target users. These OSS tools for e-learning need to be easily understood and used. The suggested enhancements aim to optimize personalization, reflecting a trend toward more sophisticated user-centered design practices, which are essential for effectively adopting these OSS projects in educational environments. The evaluation of the LMS Desktop Assistant using the Direct Observation technique has provided valuable insights into actual user

interaction, emphasizing the importance of adapting the software to the needs and expectations of end users.

In future work, we propose conducting surveys in secondary education institutions in our locality to identify the OSS e-learning tools being used. We also plan to evaluate the usability of some of these OSS tools and implement proposals for improvement. Further research is required to explore these areas and provide greater insight into the effective adoption and usability of OSS e-learning tools.

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Data availability The data collected during the study are reasonably available from the corresponding author upon request.

Declarations

Conflict of interest All authors declare no conflict of interest.

Ethical approval All participants provided informed consent before participating in the research.

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Appendix A: Primary Studies

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